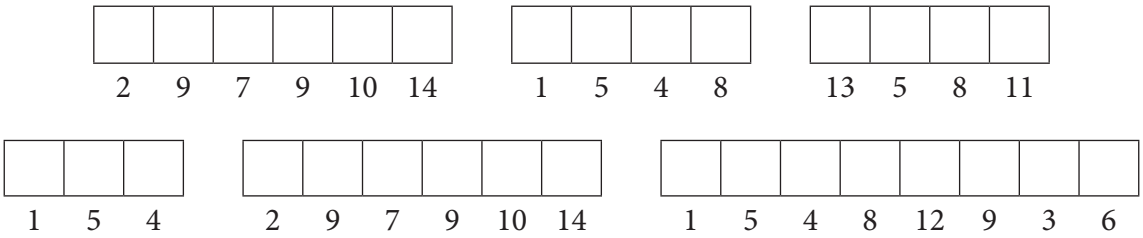
WORKSHEET

Chapter 8: Differential equations

If you …



Susan Duncan

Questions

1 Given that $\frac{dy}{dx} = x^2 + 3$, find an expression for y.

2 Given that $\frac{dy}{dx} = x^2 + 3$ describes the gradient of a graph, find an equation for the graph of the function if it is known that the graph goes through point (1, 2).

3 Given that $\frac{dv}{dt} = 10t + 3$ describes the acceleration of a body, in m s^{-2}, at time t s, find an expression for the velocity if it is known that at $t = 0$, $v = 0$.

4 The velocity of a stone that is thrown upward is given by $v = 10 - 2t$, in m s^{-1}. The stone is thrown from a height of 1 m. Find an expression for the height of the stone in terms of t.

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5 Using the equation found in Question **4**, find the height of the stone after 2 s.

6 Using the equation found in Question **4**, find the maximum height of the stone, and when this occurs.

7 Find the general solution to the differential equation $\frac{dy}{dx} = 10y + 3$. Write your answer in the form $y = Ae^{kx} - c$, where $A, k, c \in \boldsymbol{R}$.

8 The rate of decay of a radioactive material depends on the amount of material present. Write a differential equation in the form $\frac{dM}{dt} = f(M)$, where $M(t)$ is the mass present at time t.

9 The half-life of a radioactive material is 5000 years, and initially there was 0.35 g present. Using the differential equation from Question **8**, solve to find the equation for $M(t)$.

10 Using the differential equation for $M(t)$ from Question **9**, find after how many years there will be 0.12 g present.

11 Solve the differential equation $\frac{dh}{dt} = -10h$, where $t = 0, h = 2$.

12 Give the general solution to the differential equation $\frac{dy}{dx} = -y + 2$.

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13 A hot fire poker is taken out of the fire and placed on a rug in a cold room, which is at 12°C. After 10 min the fire poker has cooled to 50°C and after a further 5 min it is 30°C. Find the initial temperature of the fire poker, assuming that the rate of cooling is proportional to the difference between the temperatures of the fire poker and the room.

14 Solve the differential equation $\frac{dy}{dx} = -2y(x + 3)$.

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Solutions

A 7722 years

D $y = \frac{x^3}{3} + 3x - \frac{4}{3}$

E $M(t) \approx 0.35e^{-0.000\,1386t}$

F 26 m at $t = 5$ s

K $h = 2e^{-10t}$

L $v(t) = 5t^2 + 3t$

M $y = Ae^{10x} - 0.3$

N $y = Ae^{-x^2-6x}$

O 17 m

R $\frac{dM}{dt} = -kM$

S $y = Ae^{-x} + 2$

U $h(t) = 10t - t^2 + 1$

W 181.4°C

Y $y = \frac{x^3}{3} + 3x + c$

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